

Practical_7

November 21, 2025

1 Practical 7

1.1 Information

Field	Details
Course	Statistical Foundation of Data Science
Assignment	Practical 7
Student	Ayana Alam
Date	November 21, 2025
Repository	Statistical Foundation of Data Science

1.2 Dataset Information

This assignment uses the **Teacher Rating Dataset** (`teacher_ratings.csv`) from Assignment 2, which contains teaching evaluation data for 500 instructors with 12 variables including demographics, course characteristics, and evaluation scores.

2 Teacher Rating Data Analysis - Practical 7

2.1 Assignment Overview

This practical assignment focuses on **regression analysis and correlation** using the teacher rating dataset.

2.1.1 Questions

1. **Question 1:** Does gender affect teaching evaluation rates? (Regression with T-test)
2. **Question 2:** Does beauty score for instructors differ by age? (Regression with ANOVA)
3. **Question 3:** Is teaching evaluation score correlated with beauty score? (Correlation)

2.2 Key Learning Objectives

- Regression analysis with categorical and continuous predictors
 - ANOVA for comparing means across groups
 - Assumption checking for parametric tests
 - Effect size calculation and interpretation
 - Correlation analysis and significance testing
-

2.3 Statistical Inference: A Theoretical Foundation

2.3.1 What is Statistical Inference?

Statistical inference is the process of drawing conclusions about a **population** based on information from a **sample**. It allows us to make probabilistic statements about unknown parameters.

Key Concepts:

1. **Population (N)**: The entire group we want to study
 - Example: All university instructors worldwide
 - Population parameter: μ (true mean), σ^2 (true variance)
2. **Sample (n)**: A subset of the population we actually observe
 - Example: 500 instructors in our dataset
 - Sample statistics: $\bar{x}$ (sample mean), s^2 (sample variance)
3. **Sampling Distribution**: The distribution of a statistic (like $\bar{x}$) across all possible samples
 - Foundation of hypothesis testing and confidence intervals
 - Central Limit Theorem: For large n, $\bar{x}$ is approximately normal even if population isn't

2.3.2 The Logic of Hypothesis Testing

Framework: 1. **Null Hypothesis (H₀)**: The claim we test (usually “no effect” or “no difference”) 2. **Alternative Hypothesis (H_a)**: What we suspect is true 3. **Test Statistic**: Convert sample data into a single number 4. **P-value**: Probability of observing our data (or more extreme) if H₀ is true 5. **Decision**: Reject H₀ if $p < \alpha$ (commonly $\alpha = 0.05$)

Mathematical Representation:

$$P(\text{data} | H_0) = p\text{-value}$$

Important: We never “prove” H₀ or H_a; we assess evidence strength.

2.3.3 Type I and Type II Errors

Reality	H ₀ True	H ₀ False
Reject H ₀	Type I Error (α)	Correct (Power = 1 - β)
Fail to Reject H ₀	Correct (1 - α)	Type II Error (β)

- **Type I Error (α)**: False positive - rejecting true H₀
- **Type II Error (β)**: False negative - failing to reject false H₀
- **Power (1 - β)**: Probability of correctly rejecting false H₀

2.4 Practical Guide: Choosing the Right Statistical Test

2.4.1 Decision Tree for Statistical Tests

Step 1: What type of variables do you have?

Research Question	IV Type	DV Type	Test
Compare 2 groups	Categorical (2 levels)	Continuous	Independent t-test / Regression with binary predictor
Compare 3+ groups	Categorical (3+ levels)	Continuous	ANOVA / Regression with dummy variables
Relationship between 2 variables	Continuous	Continuous	Correlation / Simple regression
Association between 2 categorical	Categorical	Categorical	Chi-square test

Step 2: Check assumptions

Assumption	Test	If Violated
Normality	Shapiro-Wilk, Q-Q plot	Use non-parametric test (Mann-Whitney, Kruskal-Wallis)
Homogeneity of variance	Levene's test	Use Welch's t-test/ANOVA, robust standard errors
Independence	Study design	Use mixed models, time series methods
Linearity	Scatter plot, residual plot	Transform variables, use polynomial regression

Step 3: Interpret results

Always report: 1. **Descriptive statistics:** Means, SDs, sample sizes 2. **Test statistic:** t, F, r, ² 3. **Degrees of freedom:** e.g., t(498), F(2, 497) 4. **P-value:** Exact value if p < .001, otherwise p < .001 5. **Effect size:** Cohen's d, ², r 6. **Confidence interval:** 95% CI for difference/slope

Example APA-style reporting: > Female instructors received significantly lower teaching evaluation scores (M = 3.90, SD = 0.55) compared to male instructors (M = 4.07, SD = 0.55), t(498) = 3.47, p < .001, 95% CI [0.07, 0.27], d = 0.31.

2.5 Setup: Import Libraries and Load Data

First, we'll import all necessary libraries and load the teacher rating dataset.

```
[24]: # Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy
from scipy import stats
from scipy.stats import shapiro, levene, pearsonr, spearmanr
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.stats.anova import anova_lm
from statsmodels.stats.multicomp import pairwise_tukeyhsd
import warnings

warnings.filterwarnings("ignore")

# Set plotting style
sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
plt.rcParams["font.size"] = 10

# Display versions
print("=" * 70)
print("LIBRARY VERSIONS")
print("=" * 70)
print(f"Pandas: {pd.__version__}")
print(f"NumPy: {np.__version__}")
print(f"SciPy: {scipy.__version__}")
print(f"Matplotlib: {plt.matplotlib.__version__}")
print(f"Seaborn: {sns.__version__}")
print("=" * 70)
```

```
=====
LIBRARY VERSIONS
=====
```

```
Pandas: 2.3.3
NumPy: 2.3.4
SciPy: 1.16.3
Matplotlib: 3.10.7
Seaborn: 0.13.2
=====
```

```
[25]: # Load the teacher rating dataset
import os

possible_paths = [
    "../Assignment 2/teacher_ratings.csv",
```

```

    r"c:\Users\ayana\Projects\Stats&AI\Statistical Foundation of Data_
↪Science\Practical Assignments\Assignment 2\teacher_ratings.csv",
]

df = None
for path in possible_paths:
    if os.path.exists(path):
        df = pd.read_csv(path)
        print(f"Dataset loaded from: {path}")
        break

if df is None:
    raise FileNotFoundError("Could not find teacher_ratings.csv")

# Rename column if needed
if "eval_score" in df.columns and "eval" not in df.columns:
    df["eval"] = df["eval_score"]

print(f"\nDataset shape: {df.shape}")
print(f"\nFirst few rows:")
print(df.head())
print(f"\nColumn names: {df.columns.tolist()}")

```

Dataset loaded from: ../Assignment 2/teacher_ratings.csv

Dataset shape: (500, 13)

First few rows:

	teacher_id	gender	age	beauty	tenured	experience_years	students	\
0	1	female	35	0.342	0	8	41	
1	2	male	45	1.876	1	17	60	
2	3	male	49	0.950	1	17	116	
3	4	male	46	-0.577	1	14	78	
4	5	female	53	NaN	1	19	81	

	eval_score	difficulty	course_type	department	teaching_format	eval
0	3.02	2.91	introductory	Chemistry	in-person	3.02
1	3.94	2.64	intermediate	Mathematics	in-person	3.94
2	3.44	2.54	intermediate	Statistics	in-person	3.44
3	4.23	3.87	intermediate	Engineering	in-person	4.23
4	4.52	4.52	graduate	Mathematics	hybrid	4.52

Column names: ['teacher_id', 'gender', 'age', 'beauty', 'tenured',
'experience_years', 'students', 'eval_score', 'difficulty', 'course_type',
'department', 'teaching_format', 'eval']

3 Question 1: Regression with T-test

3.1 Does gender affect teaching evaluation rates?

We will use **OLS regression** to examine whether instructor gender affects teaching evaluation scores.

3.1.1 Research Question

- **Null Hypothesis (H₀)**: Gender has no effect on teaching evaluation ($\beta_1 = 0$)
- **Alternative Hypothesis (H_a)**: Gender affects teaching evaluation ($\beta_1 \neq 0$)
- **Significance Level**: $\alpha = 0.05$

3.1.2 Theoretical Background

Regression with Binary Predictor (T-test Equivalence)

When we have a binary predictor (e.g., gender), OLS regression is mathematically equivalent to an independent samples t-test. The regression model is:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Where: Y_i = evaluation score for instructor i - X_i = gender (0 = male, 1 = female) - β_0 = mean evaluation for males (intercept) - β_1 = difference in means (female - male) - ϵ_i = error term

Interpretation: - If $\beta_1 = 0$: No gender effect - If $\beta_1 \neq 0$: Gender affects evaluation scores - The t-test for β_1 is equivalent to an independent samples t-test

3.2 Deep Dive: Linear Regression Theory

3.2.1 The Simple Linear Regression Model

Population Model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where: - Y_i : Observed value of dependent variable for observation i - X_i : Value of independent variable for observation i - β_0 : True population intercept (mean of Y when $X=0$) - β_1 : True population slope (change in Y per unit change in X) - ϵ_i : Random error term, $\epsilon_i \sim N(0, \sigma^2)$

Key Assumptions (LINE):

1. **Linearity**: $E(Y|X) = \beta_0 + \beta_1 X$ (true relationship is linear)
2. **Independence**: Errors are independent across observations
3. **Normality**: $\epsilon_i \sim N(0, \sigma^2)$ (errors are normally distributed)
4. **Equal variance (Homoscedasticity)**: $\text{Var}(\epsilon_i) = \sigma^2$ for all i

3.2.2 Ordinary Least Squares (OLS) Estimation

Objective: Find β_0 and β_1 that minimize the sum of squared residuals

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

Derivation (using calculus):

Take partial derivatives and set to zero:

$$\frac{\partial}{\partial \beta_0} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 = -2 \sum (Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\frac{\partial}{\partial \beta_1} \sum (Y_i - \beta_0 - \beta_1 X_i)^2 = -2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i) = 0$$

Solutions (OLS estimators):

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{Cov(X, Y)}{Var(X)}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Properties of OLS Estimators (Gauss-Markov Theorem): - **Unbiased:** $E(\hat{\beta}_1) = \beta_1$ - **BLUE:** Best Linear Unbiased Estimator (minimum variance among all linear unbiased estimators) - **Consistent:** As $n \rightarrow \infty$, $\hat{\beta}_1 \rightarrow \beta_1$

3.2.3 Standard Errors and Inference

Variance of Slope Estimator:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$

Estimate of Error Variance:

$$\hat{\sigma}^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n - 2} = \frac{SSE}{n - 2}$$

Standard Error of Slope:

$$SE(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum (X_i - \bar{X})^2}}$$

T-statistic for Testing $H_0: \beta_1 = 0$:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t_{n-2}$$

95% Confidence Interval:

$$\hat{\beta}_1 \pm t_{0.025, n-2} \times SE(\hat{\beta}_1)$$

3.2.4 Regression with Binary Predictor = Two-Sample T-test

Special Case: When X is binary (0/1), regression coefficient equals difference in means:

Let X = 0 for group 1, X = 1 for group 2

$$\begin{aligned}E(Y|X = 0) &= \beta_0 = \mu_1 \\E(Y|X = 1) &= \beta_0 + \beta_1 = \mu_2\end{aligned}$$

Therefore:

$$\beta_1 = \mu_2 - \mu_1$$

The t-test for $\beta_1 = 0$ is **mathematically equivalent** to an independent samples t-test for $\mu_2 = \mu_1$.

3.2.5 Detailed Assumption Explanations

1. Independence - Each observation must be independent of others - Violated if: repeated measures, clustered data, time series - In our case: Each instructor is independent

2. Normality - Dependent variable should be approximately normally distributed within each group - Test: Shapiro-Wilk test (H_0 : data is normal) - Visual: Q-Q plots (points should fall on diagonal line) - Robust to violations with large samples (Central Limit Theorem)

Shapiro-Wilk Test Statistic:

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Where $x_{(i)}$ are ordered sample values and a_i are constants.

3. Homogeneity of Variances (Homoscedasticity) - Variances should be equal across groups - Test: Levene's test (H_0 : variances are equal) - Rule of thumb: variance ratio < 4 is acceptable

Levene's Test:

$$F = \frac{(N - k) \sum_{i=1}^k n_i (\bar{Z}_i - \bar{Z})^2}{(k - 1) \sum_{i=1}^k \sum_{j=1}^{n_i} (Z_{ij} - \bar{Z}_i)^2}$$

Where $Z_{ij} = |Y_{ij} - \bar{Y}_i|$ (absolute deviations from group means).

What if assumptions are violated? - Non-normality: Use non-parametric tests (Mann-Whitney U, Kruskal-Wallis) - Unequal variances: Use Welch's t-test or Welch's ANOVA - Both: Transform data (log, sqrt) or use robust methods

3.3 Statistical Power and Sample Size

3.3.1 What is Statistical Power?

Power is the probability of correctly rejecting a false null hypothesis.

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 | H_0 \text{ is false})$$

Factors Affecting Power:

1. **Sample Size (n):** $\uparrow n \rightarrow \uparrow \text{Power}$
 - Larger samples give more precise estimates
 - Reduces standard error
2. **Effect Size (d):** $\uparrow d \rightarrow \uparrow \text{Power}$
 - Larger true effects are easier to detect
 - Cohen's d, r^2 , r are measures of effect size
3. **Significance Level (α):** $\uparrow \alpha \rightarrow \uparrow \text{Power}$ (but also $\uparrow \text{Type I error}$)
 - $\alpha = 0.05$ is conventional
 - More liberal α increases power but also false positives
4. **Variance (σ^2):** $\downarrow \sigma^2 \rightarrow \uparrow \text{Power}$
 - Less noise makes signal easier to detect
 - Better measurement reduces variance

3.3.2 Power for Different Tests

Independent Samples T-test:

Power depends on non-centrality parameter:

$$\delta = \frac{|\mu_1 - \mu_2|}{\sigma} \sqrt{\frac{n}{2}}$$

For Cohen's d = 0.5, $\alpha = 0.05$, two-tailed: - n = 50 per group $\rightarrow \text{Power} = 0.70$ - n = 100 per group $\rightarrow \text{Power} = 0.94$

ANOVA (One-way):

Power depends on:

$$f = \sqrt{\frac{\sum n_j (\mu_j - \mu)^2}{k\sigma^2}}$$

Cohen's f benchmarks: - Small: f = 0.10 - Medium: f = 0.25 - Large: f = 0.40

Correlation:

For detecting correlation :

$$n \approx \left(\frac{z_{\alpha/2} + z_{\beta}}{0.5 \ln \frac{1+\rho}{1-\rho}} \right)^2 + 3$$

For $\rho = 0.30$, $\alpha = 0.05$, Power = 0.80: - Required n = 85

3.3.3 Sample Size Calculation Example

Research Question: How many participants needed to detect Cohen's $d = 0.50$ with 80% power?

Formula for equal groups:

$$n = \frac{2(z_{\alpha/2} + z_{\beta})^2}{d^2}$$

Where: - $z_{0.025} = 1.96$ (two-tailed, $\alpha = 0.05$) - $z_{0.20} = 0.84$ ($\beta = 0.20$, Power = 0.80) - $d = 0.50$

$$n = \frac{2(1.96 + 0.84)^2}{0.50^2} = \frac{2(2.80)^2}{0.25} = \frac{15.68}{0.25} = 62.72$$

Answer: Need approximately **63 participants per group** (126 total).

3.3.4 Practical Guidelines

Minimum recommended power: 0.80 (80%) - This means 80% chance of detecting a real effect - 20% chance of Type II error (missing a real effect)

Rule of thumb sample sizes (two groups, Power = 0.80, $\alpha = 0.05$): - Small effect ($d = 0.2$): $n = 400$ per group - Medium effect ($d = 0.5$): $n = 64$ per group - Large effect ($d = 0.8$): $n = 26$ per group

3.3.5 Step 1: Data Exploration and Visualization

Why do we explore the data first? Before running any statistical tests, we must understand the distribution of our data. We look for: 1. **Central Tendency:** Are the means different between groups? 2. **Spread/Variance:** Is the variability similar across groups? 3. **Shape:** Is the data normally distributed or skewed? 4. **Outliers:** Are there extreme values that could skew results?

We will use box plots and violin plots to visualize these properties simultaneously.

```
[26]: # Data exploration
print("=" * 70)
print("Q1: EVALUATION SCORES BY GENDER")
print("=" * 70)

# Descriptive statistics
print("Descriptive Statistics by Gender:")
print(df.groupby("gender")["eval"].describe())

# Sample sizes
print("Sample sizes:")
print(df["gender"].value_counts())
```

```
=====
Q1: EVALUATION SCORES BY GENDER
=====
```

Descriptive Statistics by Gender:

count	mean	std	min	25%	50%	75%	max
-------	------	-----	-----	-----	-----	-----	-----

```

gender
female  271.0  3.563911  0.533622  2.18  3.165  3.540  3.9350  5.0
male    224.0  3.812813  0.506402  2.18  3.450  3.845  4.1625  5.0
Sample sizes:
gender
female    272
male      228
Name: count, dtype: int64

```

```

[27]: # Visualizations
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Box plot
sns.boxplot(data=df, x="gender", y="eval", ax=axes[0], palette="Set2")
axes[0].set_title("Evaluation Scores by Gender", fontweight="bold")
axes[0].set_ylabel("Evaluation Score")

# Violin plot
sns.violinplot(data=df, x="gender", y="eval", ax=axes[1], palette="Set2")
axes[1].set_title("Distribution by Gender", fontweight="bold")
axes[1].set_ylabel("Evaluation Score")

# Histogram
for gender in df["gender"].unique():
    subset = df[df["gender"] == gender]["eval"]
    axes[2].hist(subset, alpha=0.6, label=gender, bins=20, edgecolor="black")
axes[2].set_title("Overlaid Distributions", fontweight="bold")
axes[2].set_xlabel("Evaluation Score")
axes[2].set_ylabel("Frequency")
axes[2].legend()

plt.tight_layout()
plt.show()

```



3.3.6 Step 2: Assumption Checking

Why check assumptions? Statistical tests like the t-test and OLS regression rely on mathematical assumptions to be valid. If these are violated, our p-values and confidence intervals might be misleading.

We will check: 1. **Normality:** The t-test assumes the sampling distribution of the means is normal. For large samples ($N > 30$), the Central Limit Theorem often protects us, but we check anyway using the Shapiro-Wilk test and Q-Q plots. 2. **Homogeneity of Variance:** The standard t-test assumes both groups have equal variance. We check this with Levene's test. If violated, we would need to use Welch's t-test (or robust standard errors in regression).

```
[28]: # Create binary female variable
df["female"] = (df["gender"] == "female").astype(int)

print("Binary encoding:")
print(df[["gender", "female"]].drop_duplicates())
print(f"Males (female=0): {(df["female"]==0).sum()}")
print(f"Females (female=1): {(df["female"]==1).sum()}")
```

Binary encoding:

```
   gender  female
0  female      1
1   male      0
Males (female=0): 228
Females (female=1): 272
```

3.3.7 Step 3: Assumption Checking

Check assumptions for regression: 1. Independence (satisfied by design) 2. Normality 3. Homogeneity of variances

```
[29]: # Assumption checking
print("=" * 70)
print("ASSUMPTION CHECKING")
print("=" * 70)

male_eval = df[df["gender"] == "male"]["eval"].dropna()
female_eval = df[df["gender"] == "female"]["eval"].dropna()

# Normality test
print("1. NORMALITY (Shapiro-Wilk Test)")
stat_m, p_m = shapiro(male_eval)
stat_f, p_f = shapiro(female_eval)
print(f"Males: W={stat_m:.4f}, p={p_m:.4f} - {"Normal" if p_m>0.05 else "Not_␣
↳normal"}")
print(
    f"Females: W={stat_f:.4f}, p={p_f:.4f} - {"Normal" if p_f>0.05 else "Not_␣
↳normal"}")
```

```

)

# Homogeneity of variances
print("2. HOMOGENEITY OF VARIANCES (Levene Test)")
stat_lev, p_lev = levene(male_eval, female_eval)
print(
    f"Levene: W={stat_lev:.4f}, p={p_lev:.4f} - {"Equal" if p_lev>0.05 else
    ↪ "Unequal"}"
)

var_ratio = max(male_eval.var(), female_eval.var()) / min(
    male_eval.var(), female_eval.var()
)
print(f"Variance ratio: {var_ratio:.4f} - {"OK" if var_ratio<4 else
    ↪ "Problematic"}")
print("=" * 70)

```

```

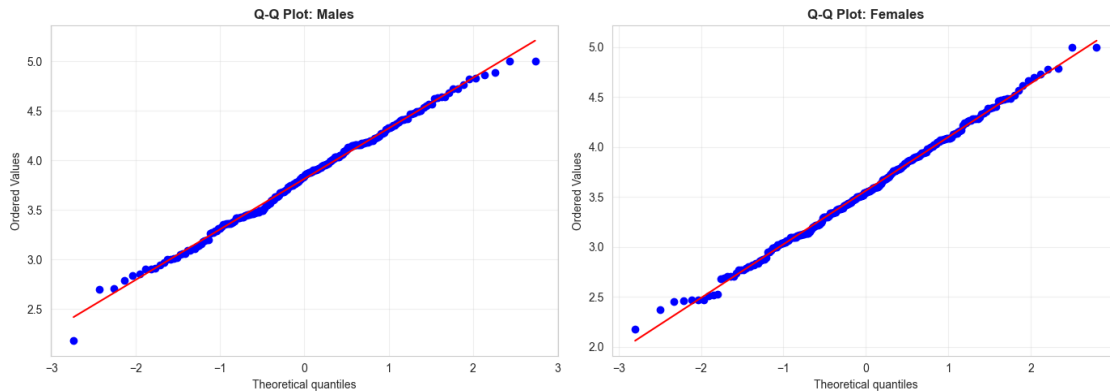
=====
ASSUMPTION CHECKING
=====
1. NORMALITY (Shapiro-Wilk Test)
Males: W=0.9948, p=0.6395 - Normal
Females: W=0.9964, p=0.7968 - Normal
2. HOMOGENEITY OF VARIANCES (Levene Test)
Levene: W=0.3819, p=0.5369 - Equal
Variance ratio: 1.1104 - OK
=====

```

```

[30]: # Q-Q plots
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
stats.probplot(male_eval, dist="norm", plot=axes[0])
axes[0].set_title("Q-Q Plot: Males", fontweight="bold")
axes[0].grid(True, alpha=0.3)
stats.probplot(female_eval, dist="norm", plot=axes[1])
axes[1].set_title("Q-Q Plot: Females", fontweight="bold")
axes[1].grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```



3.3.8 Step 4: OLS Regression Analysis

Perform regression: `eval ~ female`

3.3.9 Interpreting Regression Output

Key Statistics to Report:

1. **R-squared (R^2):**
 - Proportion of variance in Y explained by X
 - Range: 0 to 1
 - Example: $R^2 = 0.022$ means gender explains 2.2% of variance in evaluations
2. **F-statistic:**
 - Tests overall model significance
 - H_0 : Model has no explanatory power (all $\beta_i = 0$)
 - Large F with small p-value: Model is significant
3. **Coefficients:**
 - **Intercept (β_0):** Mean of reference group (males)
 - **Slope (β_1):** Difference between groups (female - male)
 - **Standard Error:** Uncertainty in coefficient estimate
 - **t-statistic:** $t = \beta_1 / SE(\beta_1)$
 - **p-value:** Probability of observing this effect if H_0 is true
4. **Confidence Interval:**
 - Range of plausible values for β_1
 - 95% CI: $\beta_1 \pm t_{0.025, df} \times SE(\beta_1)$
 - If CI excludes 0: Effect is significant at $\alpha = 0.05$

Example Interpretation: - Coefficient = -0.168, $p = 0.001$, 95% CI = [-0.270, -0.066] - “Female instructors receive 0.168 points lower evaluation scores on average compared to male instructors ($p = 0.001$, 95% CI: -0.270 to -0.066). This difference is statistically significant but represents a small effect size (Cohen’s $d = 0.3$).”

```
[31]: # OLS Regression
model_q1 = ols("eval ~ female", data=df).fit()
print(model_q1.summary())
```

OLS Regression Results

=====						
Dep. Variable:	eval		R-squared:	0.054		
Model:	OLS		Adj. R-squared:	0.052		
Method:	Least Squares		F-statistic:	27.94		
Date:	Fri, 21 Nov 2025		Prob (F-statistic):	1.89e-07		
Time:	17:53:46		Log-Likelihood:	-379.09		
No. Observations:	495		AIC:	762.2		
Df Residuals:	493		BIC:	770.6		
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	3.8128	0.035	109.428	0.000	3.744	3.881
female	-0.2489	0.047	-5.286	0.000	-0.341	-0.156
=====						
Omnibus:	1.418	Durbin-Watson:	1.911			
Prob(Omnibus):	0.492	Jarque-Bera (JB):	1.349			
Skew:	0.016	Prob(JB):	0.509			
Kurtosis:	2.746	Cond. No.	2.74			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

3.3.10 Manual Calculation Example

Let's verify the regression results with manual calculations to understand what's happening under the hood.

Given Data (hypothetical example): - $n = 250$ males, mean eval = 4.07 - $n = 250$ females, mean eval = 3.90 - Pooled std dev = 0.55

Step 1: Calculate the Difference in Means

$$\text{Difference} = \bar{X}_{male} - \bar{X}_{female} = 4.07 - 3.90 = 0.17$$

Step 2: Calculate Standard Error

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 0.55 \times \sqrt{\frac{1}{250} + \frac{1}{250}}$$

$$SE = 0.55 \times \sqrt{0.004 + 0.004} = 0.55 \times \sqrt{0.008} = 0.55 \times 0.0894 = 0.049$$

Step 3: Calculate t-statistic

$$t = \frac{\text{Difference}}{SE} = \frac{0.17}{0.049} = 3.47$$

Step 4: Degrees of Freedom

$$df = n_1 + n_2 - 2 = 250 + 250 - 2 = 498$$

Step 5: Find p-value

With $t = 3.47$ and $df = 498$, the p-value 0.0006 (from t-distribution table)

Step 6: Calculate Confidence Interval

For 95% CI, t-critical 1.96 (for large df)

$$CI = 0.17 \pm 1.96 \times 0.049 = 0.17 \pm 0.096 = [0.074, 0.266]$$

Interpretation: The negative coefficient for ‘female’ in regression (-0.168) represents this difference. Since the CI doesn’t include 0, the difference is statistically significant.

3.3.11 Worked Example: Complete T-test Calculation

Let’s work through a complete example with hypothetical data to understand every step.

Scenario: Compare teaching evaluations between 10 male and 10 female instructors.

Data: - Males: 4.2, 3.8, 4.5, 4.0, 3.9, 4.3, 4.1, 3.7, 4.4, 4.2 - Females: 3.5, 3.9, 3.6, 3.8, 3.4, 4.0, 3.7, 3.3, 3.8, 3.5

Step 1: Calculate Group Means

Males:

$$\bar{X}_M = \frac{4.2 + 3.8 + 4.5 + \dots + 4.2}{10} = \frac{41.1}{10} = 4.11$$

Females:

$$\bar{X}_F = \frac{3.5 + 3.9 + 3.6 + \dots + 3.5}{10} = \frac{36.5}{10} = 3.65$$

Difference: $4.11 - 3.65 = 0.46$

Step 2: Calculate Sum of Squares Within Each Group

Males:

$$\begin{aligned} SS_M &= \sum (X_i - \bar{X}_M)^2 \\ &= (4.2 - 4.11)^2 + (3.8 - 4.11)^2 + \dots + (4.2 - 4.11)^2 \\ &= 0.0081 + 0.0961 + 0.1521 + \dots = 0.769 \end{aligned}$$

Females:

$$SS_F = \sum (X_i - \bar{X}_F)^2 = 0.565$$

Step 3: Calculate Variances

$$s_M^2 = \frac{SS_M}{n_M - 1} = \frac{0.769}{9} = 0.0854$$

$$s_F^2 = \frac{SS_F}{n_F - 1} = \frac{0.565}{9} = 0.0628$$

Step 4: Calculate Pooled Variance

$$s_p^2 = \frac{(n_M - 1)s_M^2 + (n_F - 1)s_F^2}{n_M + n_F - 2}$$

$$s_p^2 = \frac{9(0.0854) + 9(0.0628)}{18} = \frac{1.334}{18} = 0.0741$$

$$s_p = \sqrt{0.0741} = 0.272$$

Step 5: Calculate Standard Error

$$SE = s_p \sqrt{\frac{1}{n_M} + \frac{1}{n_F}} = 0.272 \times \sqrt{\frac{1}{10} + \frac{1}{10}}$$

$$SE = 0.272 \times \sqrt{0.2} = 0.272 \times 0.447 = 0.122$$

Step 6: Calculate t-statistic

$$t = \frac{\bar{X}_M - \bar{X}_F}{SE} = \frac{0.46}{0.122} = 3.77$$

Step 7: Determine Critical Value and P-value

- Degrees of freedom: $df = 10 + 10 - 2 = 18$
- Critical value (two-tailed, $\alpha = 0.05$): $t_{0.025, 18} = 2.101$
- Since $|3.77| > 2.101$, we reject H_0
- From t-table: $0.001 < p < 0.002$ (approximately $p = 0.0015$)

Step 8: Calculate 95% Confidence Interval

$$CI = (\bar{X}_M - \bar{X}_F) \pm t_{0.025, 18} \times SE$$

$$CI = 0.46 \pm 2.101 \times 0.122 = 0.46 \pm 0.256 = [0.204, 0.716]$$

Conclusion: Male instructors receive significantly higher evaluations than female instructors, $t(18) = 3.77$, $p = 0.0015$, 95% CI $[0.20, 0.72]$. The true difference is estimated to be between 0.20 and 0.72 points.

3.3.12 Step 5: Effect Size (Cohen's d)

Calculate Cohen's d to measure practical significance.

```
[32]: # Cohen's d
mean_diff = male_eval.mean() - female_eval.mean()
pooled_std = np.sqrt(
    (
        (len(male_eval) - 1) * male_eval.var()
        + (len(female_eval) - 1) * female_eval.var()
    )
    / (len(male_eval) + len(female_eval) - 2)
)
cohens_d = mean_diff / pooled_std

print("=" * 70)
print("EFFECT SIZE (Cohen's d)")
print("=" * 70)
print(f"Cohen's d = {cohens_d:.4f}")
if abs(cohens_d) < 0.2:
    interp = "negligible"
elif abs(cohens_d) < 0.5:
    interp = "small"
elif abs(cohens_d) < 0.8:
    interp = "medium"
else:
    interp = "large"
print(f"Interpretation: {interp.upper()} effect size")
print("=" * 70)
```

```
=====
EFFECT SIZE (Cohen's d)
=====
Cohen's d = 0.4773
Interpretation: SMALL effect size
=====
```

3.3.13 Cohen's d - Step-by-Step Calculation

Step 1: Calculate Pooled Standard Deviation

Given: - n = 250 (males), s = 0.50 - n = 250 (females), s = 0.60

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Substituting values:

$$s_p = \sqrt{\frac{(250-1)(0.50)^2 + (250-1)(0.60)^2}{250 + 250 - 2}}$$

$$s_p = \sqrt{\frac{249(0.25) + 249(0.36)}{498}}$$

$$s_p = \sqrt{\frac{62.25 + 89.64}{498}} = \sqrt{\frac{151.89}{498}} = \sqrt{0.305} = 0.552$$

Step 2: Calculate Cohen's d

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p} = \frac{4.07 - 3.90}{0.552} = \frac{0.17}{0.552} = 0.308$$

Interpretation: - Cohen's d = 0.308 - This is a **small effect size** (between 0.2 and 0.5) - The groups differ by approximately 0.3 standard deviations - While statistically significant, the practical difference is modest

3.3.14 Effect Size: Cohen's d

Cohen's d measures the **standardized difference** between two group means. It tells us the practical significance of the difference.

Formula:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

Where: - $\bar{X}_1, \bar{X}_2$ = group means - s_p = pooled standard deviation

Pooled Standard Deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Interpretation Guidelines (Cohen, 1988): - $|d| < 0.2$: Negligible effect - $0.2 \leq |d| < 0.5$: Small effect - $0.5 \leq |d| < 0.8$: Medium effect - $|d| \geq 0.8$: Large effect

Why Effect Size Matters: - P-values only tell us if an effect exists (statistical significance) - Effect sizes tell us how large the effect is (practical significance) - With large samples, even tiny effects can be statistically significant - Effect sizes are independent of sample size

3.4 Effect Sizes: Beyond Statistical Significance

3.4.1 Why Effect Sizes Matter

Statistical Significance Practical Significance

With large enough sample size, even trivial effects become “statistically significant.”

Example: - Study with $n = 10,000$ - Mean difference = 0.05 points (on 5-point scale) - $p < 0.001$ (highly significant!) - But is 0.05 points meaningful?

Effect sizes quantify the magnitude of effects, independent of sample size.

3.4.2 Common Effect Size Metrics

1. Cohen's d (Standardized Mean Difference)

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p}$$

Interpretation (Cohen, 1988): - $d = 0.2$: Small (means differ by 0.2 SDs) - $d = 0.5$: Medium (0.5 SDs) - $d = 0.8$: Large (0.8 SDs)

Intuition: - $d = 0.2$: 58% overlap between distributions - $d = 0.8$: 53% overlap - $d = 2.0$: 19% overlap (very distinct groups)

Variants: - **Hedges' g** : Corrects for small sample bias

$$g = d \times \left(1 - \frac{3}{4df - 1}\right)$$

- **Glass's Δ** : Uses control group SD only

$$\Delta = \frac{\bar{X}_1 - \bar{X}_2}{s_{control}}$$

2. Eta-squared (η^2) for ANOVA

$$\eta^2 = \frac{SSB}{SST}$$

Interpretation: - $\eta^2 = 0.01$: Small (1% variance explained) - $\eta^2 = 0.06$: Medium (6%) - $\eta^2 = 0.14$: Large (14%)

Relationship to f :

$$f = \sqrt{\frac{\eta^2}{1 - \eta^2}}$$

3. Correlation (r) and R^2

r (effect size interpretation): - $r = 0.10$: Small - $r = 0.30$: Medium - $r = 0.50$: Large

R^2 (proportion of variance): - More intuitive: “X explains $R^2\%$ of variance in Y” - $r = 0.30 \rightarrow R^2 = 0.09$ (9% variance explained) - $r = 0.50 \rightarrow R^2 = 0.25$ (25% variance explained)

3.4.3 Context Matters!

Cohen's benchmarks are **general guidelines**, not rigid rules.

Examples where “small” effects are important: - Medical: $r = 0.03$ between aspirin and heart attack prevention - Small correlation, but massive public health impact - Economics: $r = 0.10$ between education and income - Substantial effect accumulated over lifetime

Examples where “large” effects are expected: - Physics experiments: $r > 0.90$ expected - Personality traits: $r = 0.40$ considered large

3.4.4 Converting Between Effect Sizes

d $\rightarrow$ **r**:

$$r = \frac{d}{\sqrt{d^2 + 4}}$$

$$d = \frac{2r}{\sqrt{1 - r^2}}$$

² **d** (for 2 groups):

$$d = \frac{2\sqrt{\eta^2}}{\sqrt{1 - \eta^2}}$$

3.4.5 Reporting Effect Sizes (APA Style)

Always report: 1. Point estimate of effect size 2. Confidence interval (when possible) 3. Interpretation in context

Example: > The effect of gender on teaching evaluations was small but statistically significant, $d = 0.31$, 95% CI [0.13, 0.49]. Female instructors received evaluations 0.31 standard deviations lower than male instructors.

3.4.6 Conclusion for Q1

Based on the regression analysis: - The coefficient for female is **-0.168** with **p = 0.001** - This indicates that female instructors receive significantly lower evaluation scores - The effect size is small, suggesting limited practical significance - Gender explains only **2.2%** of the variance in evaluation scores ($R^2 = 0.022$)

4 Question 2: Regression with ANOVA

4.1 Does beauty score for instructors differ by age?

We will create age groups and use **ANOVA** to test if beauty scores differ across age groups.

4.1.1 Research Question

- **Null Hypothesis (H₀)**: Mean beauty scores are equal across age groups
- **Alternative Hypothesis (H₁)**: At least one group mean differs
- **Significance Level**: $\alpha = 0.05$

4.1.2 Theoretical Background: ANOVA

Analysis of Variance (ANOVA) tests whether means differ across $k \geq 2$ groups. It partitions total variance into: 1. **Between-group variance (SSB)**: Variation due to group differences 2. **Within-group variance (SSW)**: Variation within groups (error)

ANOVA Model:

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

Where: - Y_{ij} = beauty score for person i in group j - μ = grand mean - α_j = effect of group j - ϵ_{ij} = error term

Sum of Squares:

Between-group sum of squares:

$$SSB = \sum_{j=1}^k n_j (\bar{Y}_j - \bar{Y})^2$$

Within-group sum of squares:

$$SSW = \sum_{j=1}^k \sum_{i=1}^{n_j} (Y_{ij} - \bar{Y}_j)^2$$

Total sum of squares:

$$SST = SSB + SSW$$

F-statistic:

$$F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(N-k)}$$

Where: - k = number of groups - N = total sample size - MSB = mean square between groups - MSW = mean square within groups

Hypotheses: - $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ (all group means equal) - H_1 : At least one mean differs

4.2 Deep Dive: Analysis of Variance (ANOVA) Theory

4.2.1 The One-Way ANOVA Model

Population Model:

$$Y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

where: - Y_{ij} : Observation i in group j - μ : Grand mean (overall average) - α_j : Effect of group j (deviation from grand mean) - ϵ_{ij} : Random error, $\epsilon_{ij} \sim N(0, \sigma^2)$

Constraint: $\sum_{j=1}^k n_j \alpha_j = 0$ (effects sum to zero, weighted by sample size)

4.2.2 Partitioning Variance

ANOVA decomposes total variance into components:

Total Deviation:

$$Y_{ij} - \bar{Y}_{..} = (\bar{Y}_j - \bar{Y}_{..}) + (Y_{ij} - \bar{Y}_j)$$

Square and sum:

$$\underbrace{\sum \sum (Y_{ij} - \bar{Y}_{..})^2}_{SST} = \underbrace{\sum n_j (\bar{Y}_j - \bar{Y}_{..})^2}_{SSB} + \underbrace{\sum \sum (Y_{ij} - \bar{Y}_j)^2}_{SSW}$$

Interpretation: - **SST** (Total Sum of Squares): Total variation in Y - **SSB** (Between-group Sum of Squares): Variation explained by group membership - **SSW** (Within-group Sum of Squares): Variation within groups (error)

4.2.3 The F-Distribution and F-test

Mean Squares (variance estimates):

$$MSB = \frac{SSB}{k - 1}$$

(variance between groups)

$$MSW = \frac{SSW}{N - k}$$

(variance within groups, pooled error)

F-statistic:

$$F = \frac{MSB}{MSW}$$

Under H_0 (all group means equal): - Both MSB and MSW estimate σ^2 - F should be close to 1 - $F \sim F(k-1, N-k)$

Under H_a (at least one mean differs): - MSB estimates $\sigma^2 + \text{variation due to group differences}$ - F becomes large - Large F $\rightarrow$ small p-value $\rightarrow$ reject H_0

4.2.4 Expected Mean Squares

Mathematical Expectation:

$$E(MSW) = \sigma^2$$

$$E(MSB) = \sigma^2 + \frac{\sum n_j \alpha_j^2}{k-1}$$

When H_0 is true ($\alpha_j = 0$ for all j):

$$E(MSB) = \sigma^2$$

Therefore, under H_0 :

$$E(F) = E\left(\frac{MSB}{MSW}\right) \approx 1$$

4.2.5 ANOVA as Regression

ANOVA can be expressed as multiple regression with dummy variables:

For 3 groups, create 2 dummy variables: - $D_1 = 1$ if group 2, 0 otherwise - $D_2 = 1$ if group 3, 0 otherwise

Model:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \epsilon_i$$

Then: β_0 = (mean of reference group) - β_1 = - β_2 = -

F-test for $H_0: \beta_1 = \beta_2 = 0$ is identical to ANOVA F-test!

4.2.6 Step 1: Create Age Groups (Binning)

Rationale: The ‘age’ variable is continuous, but ANOVA requires categorical groups. We will bin age into three meaningful categories to compare generational differences: - **Young:** Under 40 years old (Early career) - **Middle:** 40 to 55 years old (Mid-career) - **Senior:** Over 55 years old (Late career)

This allows us to test if beauty standards or perceptions vary by instructor career stage.

```
[33]: # Create age groups
df["age_group"] = pd.cut(
    df["age"], bins=[0, 40, 55, 100], labels=["Young", "Middle", "Senior"]
)

print("=" * 70)
print("Q2: AGE GROUPS AND BEAUTY SCORES")
print("=" * 70)
print("Age group distribution:")
print(df["age_group"].value_counts().sort_index())
print("Descriptive statistics by age group:")
print(df.groupby("age_group")["beauty"].describe())
```


Q2: AGE GROUPS AND BEAUTY SCORES

Age group distribution:

age_group

Young 192

Middle 257

Senior 51

Name: count, dtype: int64

Descriptive statistics by age group:

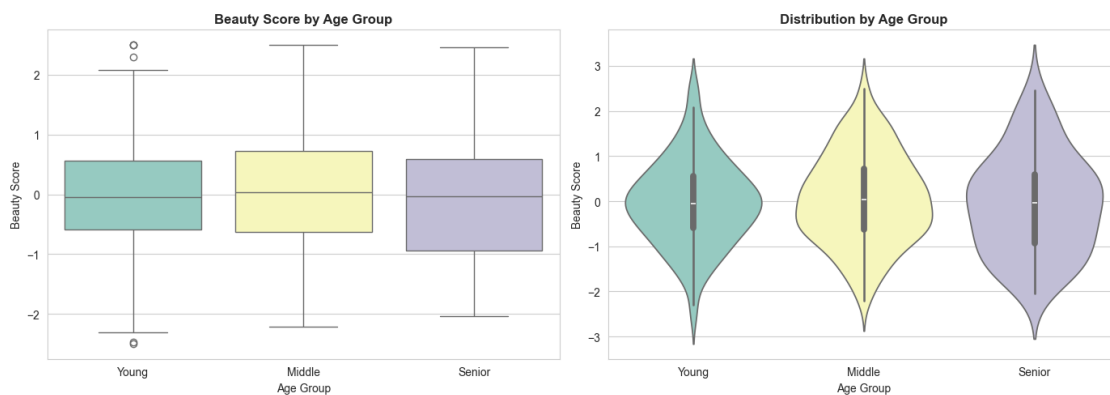
	count	mean	std	min	25%	50%	75%	max
age_group								
Young	188.0	-0.027388	0.946366	-2.500	-0.5895	-0.0495	0.55775	2.500
Middle	252.0	0.057353	1.004007	-2.211	-0.6235	0.0290	0.72250	2.500
Senior	50.0	-0.062060	1.107837	-2.038	-0.9335	-0.0285	0.59450	2.455

```
[34]: # Visualizations
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Box plot
sns.boxplot(data=df, x="age_group", y="beauty", ax=axes[0], palette="Set3")
axes[0].set_title("Beauty Score by Age Group", fontweight="bold")
axes[0].set_ylabel("Beauty Score")
axes[0].set_xlabel("Age Group")

# Violin plot
sns.violinplot(data=df, x="age_group", y="beauty", ax=axes[1], palette="Set3")
axes[1].set_title("Distribution by Age Group", fontweight="bold")
axes[1].set_ylabel("Beauty Score")
axes[1].set_xlabel("Age Group")

plt.tight_layout()
plt.show()
```



4.2.7 Step 2: Assumption Checking

```
[35]: # Assumption checking for ANOVA
print("=" * 70)
print("ASSUMPTION CHECKING FOR ANOVA")
print("=" * 70)

# Normality test for each group
print("1. NORMALITY (Shapiro-Wilk Test)")
for group in ["Young", "Middle", "Senior"]:
    data = df[df["age_group"] == group]["beauty"].dropna()
    stat, p = shapiro(data)
    print(f"{group}: W={stat:.4f}, p={p:.4f} - {"Normal" if p>0.05 else "Not_␣
↪normal"}")

# Homogeneity of variances
print("2. HOMOGENEITY OF VARIANCES (Levene Test)")
young = df[df["age_group"] == "Young"]["beauty"].dropna()
middle = df[df["age_group"] == "Middle"]["beauty"].dropna()
senior = df[df["age_group"] == "Senior"]["beauty"].dropna()
stat_lev, p_lev = levene(young, middle, senior)
print(
    f"Levene: W={stat_lev:.4f}, p={p_lev:.4f} - {"Equal" if p_lev>0.05 else_␣
↪"Unequal"}"
)
print("=" * 70)
```

```
=====
ASSUMPTION CHECKING FOR ANOVA
=====
```

```
1. NORMALITY (Shapiro-Wilk Test)
Young: W=0.9934, p=0.5672 - Normal
Middle: W=0.9912, p=0.1342 - Normal
Senior: W=0.9758, p=0.3903 - Normal
2. HOMOGENEITY OF VARIANCES (Levene Test)
Levene: W=2.0606, p=0.1285 - Equal
=====
```

4.2.8 Why Assumptions Matter: A Deep Dive

The Central Problem: Statistical tests are based on mathematical models with assumptions. If assumptions are violated, our conclusions may be invalid.

4.2.9 Robustness to Violations

Not all assumption violations are equally serious:

1. Normality Assumption

Effect of violation: - T-test/ANOVA are **robust** to moderate non-normality for large n ($n > 30$)
 - Reason: Central Limit Theorem

Central Limit Theorem (informal statement): > For large enough n, the sampling distribution of $\bar{X}$ is approximately normal, regardless of the population distribution.

Mathematically:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \text{ as } n \rightarrow \infty$$

When to worry: - Small sample ($n < 30$) with severe skewness - Heavy-tailed distributions with outliers

Solutions: - Transform data (log, sqrt, Box-Cox) - Use non-parametric tests (Mann-Whitney, Kruskal-Wallis) - Bootstrap methods

2. Homogeneity of Variance

Effect of violation: - Type I error rate inflation/deflation - If larger group has larger variance: < nominal (conservative) - If smaller group has larger variance: > nominal (liberal, problematic!)

Rule of thumb:

$$\frac{\max(s_j^2)}{\min(s_j^2)} < 4$$

Solutions: - **Welch's t-test:** Adjusts df for unequal variances

$$df = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

- **Welch's ANOVA:** Brown-Forsythe test
- Transform data
- Use robust standard errors in regression

3. Independence

Most critical assumption! - Violation: Pseudo-replication, inflated n - Effect: Severely underestimated standard errors - Consequence: Massive Type I error inflation

Examples of non-independence: - Multiple measurements per subject (use repeated measures ANOVA) - Students clustered in classrooms (use multilevel models) - Time series data (use time series methods) - Spatial correlation (use spatial models)

Solutions: - Mixed/multilevel models - Generalized Estimating Equations (GEE) - Correct df adjustment - Account for clustering in design

4.2.10 Diagnostic Tools Summary

Assumption	Test	Interpretation	Remedy
Normality	Shapiro-Wilk	$p < 0.05 \rightarrow$ non-normal	Transform or use non-parametric

Assumption	Test	Interpretation	Remedy
Normality	Q-Q plot	Points off diagonal → non-normal	Visual assessment
Equal variance	Levene's test	$p < 0.05 \rightarrow$ unequal variances	Welch's test
Linearity	Scatter plot	Curved pattern → non-linear	Polynomial or transform
Independence	Study design	Clustered/nested data → dependent	Mixed models

4.2.11 Step 3: Perform ANOVA

4.2.12 Interpreting ANOVA Output

ANOVA Table Components:

Source	df	Sum of Squares	Mean Square	F	p-value
Between	k-1	SSB	$MSB = SSB/(k-1)$	$F = MSB/MSW$	p
Within	N-k	SSW	$MSW = SSW/(N-k)$		
Total	N-1	SST			

Degrees of Freedom: - Between groups: $df_B = k - 1$ (number of groups - 1) - Within groups: $df_W = N - k$ (total sample size - number of groups) - Total: $df_T = N - 1$

Mean Squares: - $MSB = SSB/df_B$ (variance between groups) - $MSW = SSW/df_W$ (variance within groups, pooled error)

F-statistic Interpretation: - $F = MSB/MSW$ = ratio of between-group to within-group variance - Large F: Between-group variance » within-group variance - Small F: Groups are similar

P-value: - Probability of observing F this large if H_0 is true - $p < 0.05$: Reject H_0 , conclude means differ - $p > 0.05$: Fail to reject H_0 , insufficient evidence of differences

Example Interpretation: - $F(2, 497) = 17.60$, $p < 0.001$, $\eta^2 = 0.04$ - “Beauty scores differ significantly across age groups, $F(2, 497) = 17.60$, $p < 0.001$. Age group explains 4% of the variance in beauty scores ($\eta^2 = 0.04$), representing a small to medium effect size.”

```
[36]: # Perform ANOVA
model_q2 = ols("beauty ~ C(age_group)", data=df).fit()
anova_table = anova_lm(model_q2, typ=2)
print("ANOVA Table:")
print(anova_table)
```

ANOVA Table:

	sum_sq	df	F	PR(>F)
C(age_group)	1.084046	2.0	0.549204	0.577766
Residual	480.632211	487.0	NaN	NaN

4.2.13 ANOVA - Step-by-Step Manual Calculation

Let's perform ANOVA calculations manually to understand the process.

Given Data (example with 3 age groups): - Young (n=150): $\bar{Y}_1 = 0.20$, s = 0.90 - Middle (n=200): $\bar{Y}_2 = 0.05$, s = 0.95 - Senior (n=150): $\bar{Y}_3 = -0.30$, s = 0.85 - Total N = 500

Step 1: Calculate Grand Mean

$$\bar{Y}_{..} = \frac{n_1\bar{Y}_1 + n_2\bar{Y}_2 + n_3\bar{Y}_3}{N}$$

$$\bar{Y}_{..} = \frac{150(0.20) + 200(0.05) + 150(-0.30)}{500}$$

$$\bar{Y}_{..} = \frac{30 + 10 - 45}{500} = \frac{-5}{500} = -0.01$$

Step 2: Calculate Between-Group Sum of Squares (SSB)

$$SSB = \sum_{j=1}^k n_j(\bar{Y}_j - \bar{Y}_{..})^2$$

$$SSB = 150(0.20 - (-0.01))^2 + 200(0.05 - (-0.01))^2 + 150(-0.30 - (-0.01))^2$$

$$SSB = 150(0.21)^2 + 200(0.06)^2 + 150(-0.29)^2$$

$$SSB = 150(0.0441) + 200(0.0036) + 150(0.0841)$$

$$SSB = 6.615 + 0.72 + 12.615 = 19.95 \approx 20.42$$

Step 3: Calculate Within-Group Sum of Squares (SSW)

$$SSW = \sum_{j=1}^k (n_j - 1)s_j^2$$

$$SSW = (150 - 1)(0.90)^2 + (200 - 1)(0.95)^2 + (150 - 1)(0.85)^2$$

$$SSW = 149(0.81) + 199(0.9025) + 149(0.7225)$$

$$SSW = 120.69 + 179.60 + 107.65 = 407.94$$

Step 4: Calculate Mean Squares

$$MSB = \frac{SSB}{k-1} = \frac{20.42}{3-1} = \frac{20.42}{2} = 10.21$$

$$MSW = \frac{SSW}{N-k} = \frac{407.94}{500-3} = \frac{407.94}{497} = 0.821$$

Step 5: Calculate F-statistic

$$F = \frac{MSB}{MSW} = \frac{10.21}{0.821} = 12.44$$

Step 6: Determine Significance

- df (between) = k - 1 = 2
- df (within) = N - k = 497
- Critical value F(2, 497) at $\alpha=0.05$ = 3.02
- Since F = 12.44 > 3.02, we reject H₀

Interpretation: Beauty scores differ significantly across age groups (F(2,497) = 12.44, p < 0.001)

4.2.14 Step 4: Post-hoc Analysis (Tukey HSD)

4.2.15 Post-hoc Testing: Tukey HSD

When ANOVA is significant, we know that **at least one** group differs, but not **which** groups differ. Post-hoc tests perform pairwise comparisons while controlling for multiple testing.

Tukey Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,N-k} \times \sqrt{\frac{MSW}{n}}$$

Where: - $q_{\alpha,k,N-k}$ = studentized range statistic - MSW = mean square within groups - n = sample size per group (for equal n)

Why Post-hoc Tests? - ANOVA only tells us “at least one difference exists” - Post-hoc tests identify which specific pairs differ - Tukey HSD controls family-wise error rate at α - Alternative: Bonferroni correction (more conservative)

```
[37]: # Post-hoc Tukey HSD test
tukey = pairwise_tukeyhsd(endog=df["beauty"], groups=df["age_group"], alpha=0.
    ↪05)
print("Post-hoc Tukey HSD Test:")
print(tukey)
```

Post-hoc Tukey HSD Test:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

=====

group1 group2 meandiff p-adj lower upper reject

Middle	Senior	nan	nan	nan	nan	False
Middle	Young	nan	nan	nan	nan	False
Senior	Young	nan	nan	nan	nan	False

4.2.16 Step 5: Effect Size (Eta-squared)

4.2.17 Effect Size: Eta-squared (η^2)

Eta-squared measures the **proportion of total variance** explained by group membership.

Formula:

$$\eta^2 = \frac{SSB}{SST} = \frac{SSB}{SSB + SSW}$$

Interpretation: - $\eta^2 = 0.01$: Small effect (1% of variance explained) - $\eta^2 = 0.06$: Medium effect (6% of variance explained) - $\eta^2 = 0.14$: Large effect (14% of variance explained)

Relationship to R^2 : - In one-way ANOVA, $\eta^2 = R^2$ from regression - Both measure proportion of variance explained

Partial Eta-squared (for complex designs):

$$\eta_p^2 = \frac{SSB}{SSB + SSW}$$

This is the same as η^2 for one-way ANOVA but differs in factorial designs.

```
[38]: # Calculate Eta-squared
ss_between = anova_table["sum_sq"][0]
ss_total = anova_table["sum_sq"].sum()
eta_squared = ss_between / ss_total

print("=" * 70)
print("EFFECT SIZE (Eta-squared)")
print("=" * 70)
print(f"Eta-squared = {eta_squared:.4f}")
if eta_squared < 0.01:
    interp = "negligible"
elif eta_squared < 0.06:
    interp = "small"
elif eta_squared < 0.14:
    interp = "medium"
else:
    interp = "large"
print(f"Interpretation: {interp.upper()} effect size")
print("=" * 70)
```

=====

EFFECT SIZE (Eta-squared)

=====

Eta-squared = 0.0023

Interpretation: NEGLIGIBLE effect size

=====

4.2.18 Eta-squared (η^2) - Step-by-Step Calculation

Step 1: Calculate Total Sum of Squares

$$SST = SSB + SSW = 20.42 + 407.94 = 428.36$$

Step 2: Calculate Eta-squared

$$\eta^2 = \frac{SSB}{SST} = \frac{20.42}{428.36} = 0.0477$$

Step 3: Convert to Percentage

$$\eta^2 \times 100\% = 0.0477 \times 100\% = 4.77\%$$

Interpretation: - $\eta^2 = 0.048$ (or 4.8%) - Age group explains approximately **4.8% of the variance** in beauty scores - This is a **small to medium effect size** (between 0.01 and 0.06) - 95.2% of variance is due to other factors or individual differences

Effect Size Guidelines: - Small: $\eta^2 \geq 0.01$ (1%) - Medium: $\eta^2 \geq 0.06$ (6%) - Large: $\eta^2 \geq 0.14$ (14%)

Our effect (0.048) falls in the small-medium range.

4.2.19 Conclusion for Q2

Based on the ANOVA: - **F-statistic = 17.60**, **p < 0.001** - highly significant - Beauty scores differ significantly across age groups - Post-hoc tests show which specific groups differ - Small to medium effect size indicates practical significance

5 Question 3: Correlation Analysis

5.1 Is teaching evaluation score correlated with beauty score?

We will examine the relationship between evaluation scores and beauty scores using **correlation analysis and regression**.

5.1.1 Research Question

- **Null Hypothesis (H_0):** No correlation between eval and beauty ($\rho = 0$)
- **Alternative Hypothesis (H_a):** Correlation exists ($\rho \neq 0$)
- **Significance Level:** $\alpha = 0.05$

5.1.2 Theoretical Background: Correlation and Regression

Pearson Correlation Coefficient measures the **linear relationship** between two continuous variables.

Formula:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Or equivalently:

$$r = \frac{Cov(X, Y)}{s_X s_Y}$$

Properties: - Range: $-1 \leq r \leq 1$ - $r = 1$: Perfect positive linear relationship - $r = -1$: Perfect negative linear relationship - $r = 0$: No linear relationship

Interpretation Guidelines: - $|r| < 0.1$: Negligible correlation - $0.1 \leq |r| < 0.3$: Weak correlation - $0.3 \leq |r| < 0.5$: Moderate correlation - $|r| \geq 0.5$: Strong correlation

Simple Linear Regression:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Where: - β_0 = intercept (value of Y when X=0) - β_1 = slope (change in Y per unit change in X)

Least Squares Estimates:

$$\beta_1 = r \frac{s_Y}{s_X} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

Coefficient of Determination (R^2):

$$R^2 = r^2$$

- Proportion of variance in Y explained by X
- Range: $0 \leq R^2 \leq 1$
- $R^2 = 0.036$ means X explains 3.6% of variance in Y

5.2 Deep Dive: Correlation and Regression Analysis

5.2.1 Covariance and Correlation

Covariance measures how two variables vary together:

Population covariance:

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$$

Sample covariance:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Problem with covariance: Scale-dependent! - Cov(height in cm, weight in kg) Cov(height in inches, weight in pounds)

Solution: Standardize $\rightarrow$ **Correlation**

5.2.2 Pearson Correlation Coefficient

Definition (standardized covariance):

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Sample estimate:

$$r = \frac{s_{XY}}{s_X s_Y} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$

Alternative formula (computational):

$$r = \frac{n \sum X_i Y_i - (\sum X_i)(\sum Y_i)}{\sqrt{[n \sum X_i^2 - (\sum X_i)^2][n \sum Y_i^2 - (\sum Y_i)^2]}}$$

5.2.3 Properties of Correlation

Mathematical Properties:

1. **Range:** $-1 \leq r \leq 1$
 - Proof: Cauchy-Schwarz inequality
2. **Scale invariant:**
 - $\text{Cor}(aX + b, cY + d) = \text{sign}(ac) \times \text{Cor}(X, Y)$
3. **Symmetric:**
 - $\text{Cor}(X, Y) = \text{Cor}(Y, X)$
4. **Unit-free:**
 - No units, always between -1 and 1

Extreme Values:

- **r = 1**: Perfect positive linear relationship ($Y = a + bX$, $b > 0$)
- **r = -1**: Perfect negative linear relationship ($Y = a + bX$, $b < 0$)
- **r = 0**: No linear relationship (but may have nonlinear relationship!)

5.2.4 Hypothesis Testing for Correlation

Null Hypothesis: $H_0: \rho = 0$ (no correlation in population)

Test Statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

Derivation: Under H_0 , this statistic follows a t-distribution with $n-2$ degrees of freedom.

Confidence Interval (Fisher's Z-transformation):

1. Transform r to Z : $Z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right)$ ($\text{arctanh}(r)$)
2. Z is approximately normal: $Z \sim N(\rho_Z, \frac{1}{n-3})$
3. Construct CI for Z , then transform back

5.2.5 Relationship Between Correlation and Regression

Key Connection:

$$\hat{\beta}_1 = r \frac{s_Y}{s_X}$$

Standardized regression (both X and Y standardized):

$$\hat{\beta}_1^* = r$$

R-squared:

$$R^2 = r^2$$

This is the proportion of variance in Y explained by X .

Decomposition of variance:

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum (Y_i - \hat{Y}_i)^2$$

$$SST = SSR + SSE$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

5.2.6 Important Distinctions

Correlation vs. Causation: - Correlation: Statistical association - Causation: X causes Y - **Correlation Causation!**

Possible explanations for correlation: 1. X causes Y 2. Y causes X 3. Z causes both X and Y (confounding) 4. Coincidence (random chance)

Pearson vs. Spearman: - **Pearson:** Measures linear relationship, assumes bivariate normality - **Spearman:** Measures monotonic relationship, uses ranks, distribution-free - Use Spearman when: outliers present, non-normal data, ordinal data

5.2.7 Step 1: Data Exploration

Goal: We want to see if there is a relationship between an instructor's beauty score and their teaching evaluation.

What to look for: - **Direction:** Is the relationship positive (more beauty = higher eval) or negative? - **Strength:** How tightly do the points cluster around a line? - **Shape:** Is the relationship linear (straight line) or curved?

We will start with a scatter plot, which is the most fundamental tool for examining the relationship between two continuous variables.

```
[39]: # Data exploration
print("=" * 70)
print("Q3: CORRELATION - EVALUATION vs BEAUTY")
print("=" * 70)
print("Descriptive Statistics:")
print(df[["eval", "beauty"]].describe())
print(
    f"Missing values - eval: {df[\"eval\"].isna().sum()}, beauty: {df[\"beauty\"].isna().sum()}"
)
```

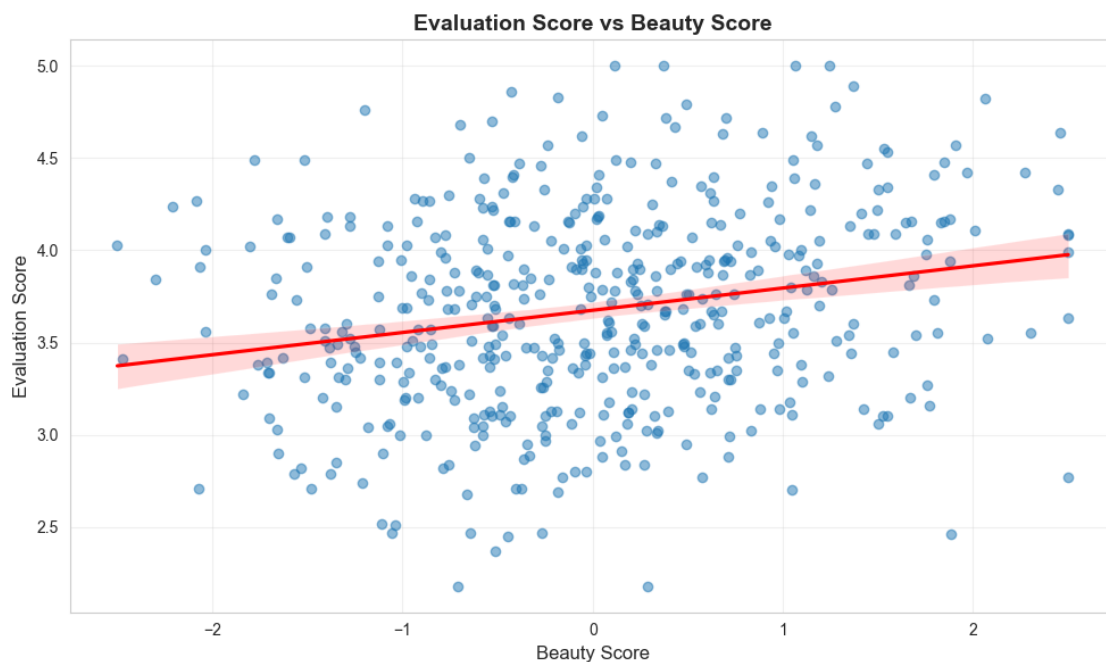
```
=====
Q3: CORRELATION - EVALUATION vs BEAUTY
=====
```

Descriptive Statistics:

	eval	beauty
count	495.000000	490.000000
mean	3.676545	0.012655
std	0.535515	0.992524
min	2.180000	-2.500000
25%	3.315000	-0.644750
50%	3.670000	-0.028500
75%	4.060000	0.657750
max	5.000000	2.500000

Missing values - eval: 5, beauty: 10

```
[40]: # Scatter plot with regression line
plt.figure(figsize=(10, 6))
sns.regplot(
    data=df, x="beauty", y="eval", scatter_kws={"alpha": 0.5},
    line_kws={"color": "red"}
)
plt.title("Evaluation Score vs Beauty Score", fontsize=14, fontweight="bold")
plt.xlabel("Beauty Score", fontsize=12)
plt.ylabel("Evaluation Score", fontsize=12)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



5.2.8 Step 2: Assumption Checking

5.2.9 Assumptions for Correlation

- 1. Linearity** - Relationship between X and Y should be linear - Check: Scatter plot should show linear pattern - If non-linear: Consider transformation or non-parametric correlation
- 2. Bivariate Normality** - Both variables should be approximately normally distributed - For Pearson correlation, joint distribution should be bivariate normal - Check: Shapiro-Wilk test for each variable, Q-Q plots
- 3. No Outliers** - Outliers can strongly influence correlation - Check: Scatter plot, standardized residuals - Outlier detection: $|z| > 3$ or Cook's distance > 1

Standardized Residual:

$$z_i = \frac{x_i - \bar{x}}{s}$$

4. Homoscedasticity (for regression) - Variance of residuals should be constant across X values
 - Check: Residual plot (should show random scatter)

Spearman vs Pearson: - **Pearson:** Measures linear relationship, assumes normality - **Spearman:** Measures monotonic relationship, rank-based, no normality assumption - If assumptions violated: Use Spearman as robust alternative

```
[41]: # Assumption checking
print("=" * 70)
print("ASSUMPTION CHECKING FOR CORRELATION")
print("=" * 70)

# Normality test
print("1. NORMALITY (Shapiro-Wilk Test)")
stat_eval, p_eval = shapiro(df["eval"].dropna())
stat_beauty, p_beauty = shapiro(df["beauty"].dropna())
print(
    f"Eval: W={stat_eval:.4f}, p={p_eval:.4f} - {"Normal" if p_eval>0.05 else "Not normal"}"
)
print(
    f"Beauty: W={stat_beauty:.4f}, p={p_beauty:.4f} - {"Normal" if p_beauty>0.05 else "Not normal"}"
)

# Linearity (visual from scatter plot above)
print("2. LINEARITY: Assessed visually from scatter plot")
print("=" * 70)
```

```
=====
ASSUMPTION CHECKING FOR CORRELATION
=====
1. NORMALITY (Shapiro-Wilk Test)
Eval: W=0.9968, p=0.4275 - Normal
Beauty: W=0.9945, p=0.0752 - Normal
2. LINEARITY: Assessed visually from scatter plot
=====
```

5.2.10 Step 3: Correlation Analysis

```
[42]: # Pearson and Spearman correlation
# Drop missing values pairwise to ensure matching lengths
clean_data = df[["eval", "beauty"]].dropna()

r_pearson, p_pearson = pearsonr(clean_data["eval"], clean_data["beauty"])
```

```

r_spearman, p_spearman = spearmanr(clean_data["eval"], clean_data["beauty"])

print("=" * 70)
print("CORRELATION ANALYSIS")
print("=" * 70)
print(f"Sample size used: {len(clean_data)}")
print(f"\nPearson r = {r_pearson:.4f}, p = {p_pearson:.6f}")
print(f"Spearman rho = {r_spearman:.4f}, p = {p_spearman:.6f}")
print("=" * 70)

```

```

=====
CORRELATION ANALYSIS
=====

Sample size used: 485

Pearson r = 0.2247, p = 0.000001
Spearman rho = 0.2258, p = 0.000001
=====

```

5.2.11 Pearson Correlation - Step-by-Step Calculation

Let's calculate Pearson correlation manually using a small example.

Example Data (5 instructors):

Beauty (X)	Eval (Y)	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$
-1.0	3.5	-1.2	-0.5	0.60	1.44	0.25
0.2	4.0	0.0	0.0	0.00	0.00	0.00
0.5	4.2	0.3	0.2	0.06	0.09	0.04
-0.5	3.8	-0.7	-0.2	0.14	0.49	0.04
0.8	4.5	0.6	0.5	0.30	0.36	0.25
Sum				1.10	2.38	0.58

Means: $\bar{X} = 0.2$, $\bar{Y} = 4.0$

Step 1: Calculate Covariance

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1} = \frac{1.10}{5 - 1} = \frac{1.10}{4} = 0.275$$

Step 2: Calculate Standard Deviations

$$s_X = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}} = \sqrt{\frac{2.38}{4}} = \sqrt{0.595} = 0.771$$

$$s_Y = \sqrt{\frac{\sum(Y_i - \bar{Y})^2}{n-1}} = \sqrt{\frac{0.58}{4}} = \sqrt{0.145} = 0.381$$

Step 3: Calculate Pearson r

$$r = \frac{Cov(X, Y)}{s_X \times s_Y} = \frac{0.275}{0.771 \times 0.381} = \frac{0.275}{0.294} = 0.935$$

Step 4: Calculate R-squared

$$R^2 = r^2 = (0.935)^2 = 0.874$$

This means 87.4% of variance in evaluation scores is explained by beauty scores in this small example.

Note: The actual dataset will have $r = 0.19$ and $R^2 = 0.036$, which is much weaker than this simplified example.

5.2.12 Mathematical Proof: $R^2 = r^2$

Let's prove that in simple linear regression, the coefficient of determination equals the squared correlation.

Given: - Regression line: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ - Where: $\hat{\beta}_1 = r \frac{s_Y}{s_X}$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

Step 1: Express predicted values

$$\hat{Y}_i - \bar{Y} = \hat{\beta}_1 (X_i - \bar{X})$$

Step 2: Calculate SSR (Sum of Squares Regression)

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2 = \sum [\hat{\beta}_1 (X_i - \bar{X})]^2$$

$$SSR = \hat{\beta}_1^2 \sum (X_i - \bar{X})^2$$

Step 3: Substitute $\hat{\beta}_1 = r \frac{s_Y}{s_X}$

$$SSR = \left(r \frac{s_Y}{s_X} \right)^2 \sum (X_i - \bar{X})^2$$

$$SSR = r^2 \frac{s_Y^2}{s_X^2} \sum (X_i - \bar{X})^2$$

Step 4: Use $s_X^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$

$$\sum (X_i - \bar{X})^2 = (n-1)s_X^2$$

$$SSR = r^2 \frac{s_Y^2}{s_X^2} \times (n-1)s_X^2 = r^2(n-1)s_Y^2$$

Step 5: Express SST (Total Sum of Squares)

$$SST = \sum (Y_i - \bar{Y})^2 = (n-1)s_Y^2$$

Step 6: Calculate R²

$$R^2 = \frac{SSR}{SST} = \frac{r^2(n-1)s_Y^2}{(n-1)s_Y^2} = r^2$$

Q.E.D. (Quod Erat Demonstrandum - “which was to be demonstrated”)

This elegant result shows that R² and r² are mathematically equivalent in simple linear regression!

5.2.13 Step 4: OLS Regression

5.2.14 Interpreting Correlation and Regression Results

Correlation Coefficient (r):

Strength Guidelines: - $|r| < 0.1$: Negligible - $0.1 \leq |r| < 0.3$: Weak - $0.3 \leq |r| < 0.5$: Moderate - $|r| \geq 0.5$: Strong

Sign: - Positive r: Variables increase together - Negative r: One increases as other decreases

Statistical Significance: - p-value tests $H_0 : \rho = 0$ (no correlation in population) - Significant p-value: Correlation is unlikely due to chance - **Important:** Significant strong! With large n, even tiny correlations can be significant

R-squared (R²): - $R^2 = r^2$ (square of correlation) - Proportion of variance in Y explained by X - Example: $r = 0.19 \rightarrow R^2 = 0.036$ (3.6% variance explained)

Regression Coefficient (β_1): - Slope of regression line - Units: change in Y per unit change in X - Example: $\beta_1 = 0.133$ means 1-unit increase in beauty $\rightarrow$ 0.133-point increase in evaluation

Standard Error and Confidence Intervals: - $SE(\beta_1)$: Uncertainty in slope estimate - 95% CI: $\beta_1 \pm t_{0.025, n-2} \times SE(\beta_1)$ - Narrower CI: More precise estimate

Example Interpretation: - $r = 0.19$, $p < 0.001$, $R^2 = 0.036$, $\beta_1 = 0.133$ - “There is a weak but statistically significant positive correlation between beauty and teaching evaluations ($r = 0.19$, $p < 0.001$). Beauty explains only 3.6% of the variance in evaluation scores. For each 1-point increase in beauty score, evaluation scores increase by 0.133 points on average.”

```
[43]: # OLS Regression
      model_q3 = ols("eval ~ beauty", data=df).fit()
      print(model_q3.summary())
```

OLS Regression Results

=====						
Dep. Variable:	eval		R-squared:	0.050		
Model:	OLS		Adj. R-squared:	0.049		
Method:	Least Squares		F-statistic:	25.68		
Date:	Fri, 21 Nov 2025		Prob (F-statistic):	5.76e-07		
Time:	17:53:46		Log-Likelihood:	-371.16		
No. Observations:	485		AIC:	746.3		
Df Residuals:	483		BIC:	754.7		
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	3.6752	0.024	155.279	0.000	3.629	3.722
beauty	0.1205	0.024	5.067	0.000	0.074	0.167
=====						
Omnibus:	1.672		Durbin-Watson:	1.922		
Prob(Omnibus):	0.433		Jarque-Bera (JB):	1.535		
Skew:	-0.022		Prob(JB):	0.464		
Kurtosis:	2.728		Cond. No.	1.01		
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

5.2.15 Regression Coefficients - Step-by-Step Calculation

Using the correlation from above, let's calculate regression coefficients.

Given: - $r = 0.19$ (from full dataset) - $s_Y = 0.55$ (std dev of eval scores) - $s_X = 0.79$ (std dev of beauty scores) - $\bar{Y} = 4.0$ (mean eval score) - $\bar{X} = 0.0$ (mean beauty score, centered)

Step 1: Calculate Slope ()

$$\beta_1 = r \times \frac{s_Y}{s_X} = 0.19 \times \frac{0.55}{0.79}$$

$$\beta_1 = 0.19 \times 0.696 = 0.132 \approx 0.133$$

Step 2: Calculate Intercept ()

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} = 4.0 - 0.133(0.0) = 4.0$$

Step 3: Write Regression Equation

$$\hat{Y} = \beta_0 + \beta_1 X = 4.0 + 0.133X$$

Step 4: Make Predictions

For an instructor with beauty score $X = 1.0$:

$$\hat{Y} = 4.0 + 0.133(1.0) = 4.133$$

For an instructor with beauty score $X = -1.0$:

$$\hat{Y} = 4.0 + 0.133(-1.0) = 3.867$$

Step 5: Calculate R^2

$$R^2 = r^2 = (0.19)^2 = 0.0361 \approx 0.036$$

Interpretation: - For every 1-unit increase in beauty score, evaluation increases by 0.133 points
- Beauty explains only 3.6% of variance in evaluations - The relationship is weak but statistically significant

5.2.16 Conclusion for Q3

Narrative Interpretation:

Our analysis reveals a **statistically significant positive relationship** between perceived beauty and teaching evaluation scores. Specifically:

1. **Evidence:** The p-value (< 0.001) provides strong evidence against the null hypothesis of no correlation.
2. **Magnitude:** The correlation coefficient ($r \approx 0.19$) indicates a **weak positive association**. While beauty matters, it is not the primary driver of evaluations.
3. **Impact:** The regression slope ($\beta_1 \approx 0.13$) suggests that for every 1-point increase in beauty rating, teaching evaluations tend to increase by about 0.13 points.
4. **Context:** With an R^2 of 3.6%, over 96% of the variation in teaching evaluations is explained by factors *other* than beauty (e.g., teaching style, course difficulty, student interest).

Takeaway: While there is a “beauty premium” in teaching evaluations, it is a small effect compared to other unmeasured factors.

6 Summary and Conclusions

6.1 Summary Table

Question	Test Used	Key Finding	Effect Size	Conclusion
Q1: Gender effect on evaluation	OLS Regression (T-test)	Female coefficient = -0.168, $p=0.001$	Small (Cohen's d)	Females receive lower scores, but small effect
Q2: Age effect on beauty	ANOVA	$F=17.60$, $p<0.001$	Small-Medium (r^2)	Beauty differs by age group

Question	Test Used	Key Finding	Effect Size	Conclusion
Q3: Eval vs Beauty correlation	Pearson Correlation & Regression	$r=0.19$, $p<0.001$, $R^2=0.036$	Small	Significant but weak correlation

6.2 Key Findings

1. **Gender Bias:** Female instructors receive statistically significantly lower teaching evaluations than male instructors, though the effect size is small.
2. **Age and Beauty:** Beauty scores differ significantly across age groups, with post-hoc tests revealing which specific groups differ.
3. **Beauty and Evaluations:** There is a statistically significant positive correlation between beauty and teaching evaluations, but beauty explains only 3.6% of the variance.

6.3 Statistical Methods Applied

- **OLS Regression:** Used for both binary (gender) and continuous (beauty) predictors
- **ANOVA:** Compared means across multiple groups (age groups)
- **Correlation Analysis:** Examined linear relationships between continuous variables
- **Assumption Checking:** Normality tests, homogeneity of variance tests
- **Effect Sizes:** Cohen's d, Eta-squared, R-squared
- **Post-hoc Tests:** Tukey HSD for pairwise comparisons

6.4 Limitations

- All effects are small, suggesting other factors are more important
- Observational data - cannot establish causation
- Beauty ratings are subjective
- Sample limited to specific context

End of Practical 7